

Plane wave propagation

Previously...

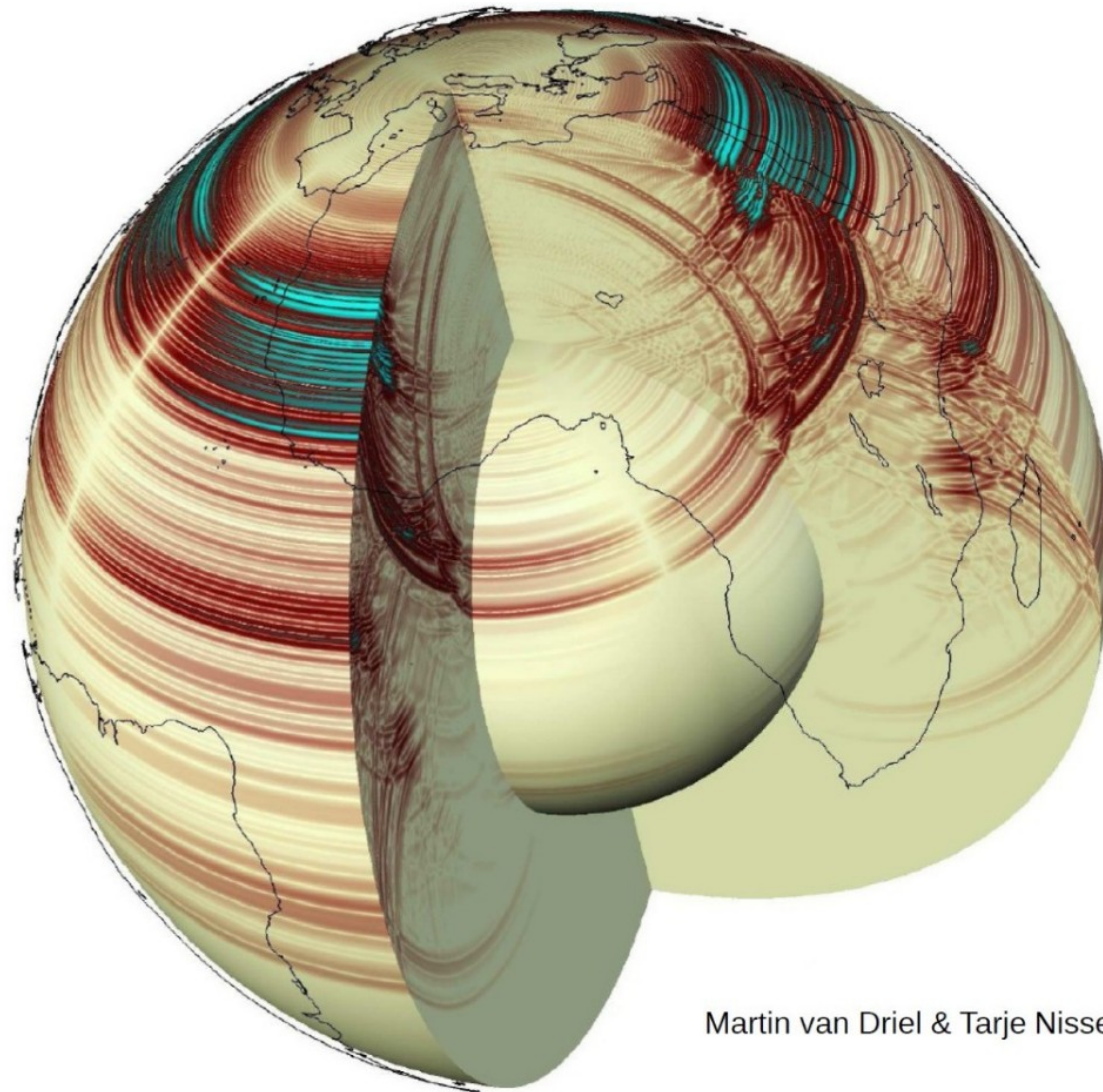
Linearised equations of motion including a stress glut:

$$\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = - \frac{\partial \bar{S}_{ij}}{\partial x_j},$$

$$A_{ijkl} \frac{\partial u_k}{\partial x_l} \hat{n}_j = \bar{S}_{ij} \hat{n}_j$$

Definition of the elastic tensor:

$$A_{ijkl}(\mathbf{x}) = \frac{\partial^2 W}{\partial F_{ij} \partial F_{kl}}(\mathbf{x}, \mathbf{I}),$$



Martin van Driel & Tarje Nissen Meyer

The Christoffel equation

Plane wave ansatz

We consider an infinite and homogeneous body

Look for a plane wave solution of the form:

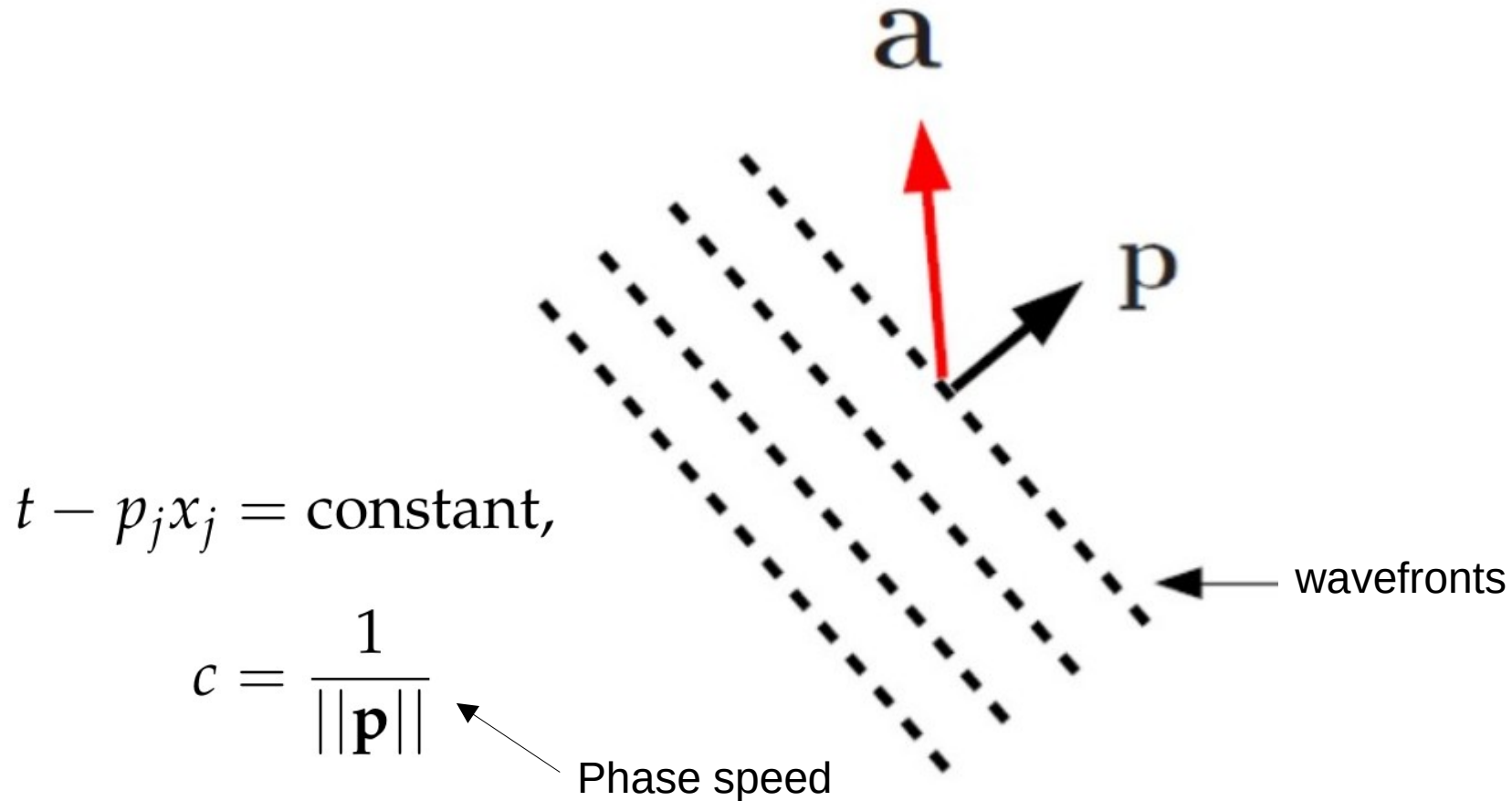
$$u_i(x, t) = f(t - p_j x_j) a_i$$

The **slowness vector**, $\mathbf{p}$, sets the orientation and speed of the wavefronts

The **polarisation vector**, $\mathbf{a}$, sets the direction of particle motion

The **pulse shape** set by f , and does not vary as the wave propagates – in this sense the waves are **non-dispersive** (but...)

Plane wave ansatz



The Christoffel equation

For the plane wave ansatz:

$$\frac{\partial^2 u_i}{\partial t^2} = f''(t - p_j x_j) a_i,$$

$$\frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = f''(t - p_m x_m) A_{ijkl} p_j p_l a_k$$

A non-trivial plane wave solution exists if and only if:

$$(\rho \delta_{ik} - A_{ijkl} p_j p_l) a_k = 0.$$

This is known as the **Christoffel equation** – it places a constraint on both the slowness and polarisation vectors

The Christoffel matrix

It will be useful to define the **Christoffel matrix**, which is a function of $\mathbf{p}$:

$$\Gamma_{ik}(\mathbf{p}) = \frac{1}{\rho} A_{ijkl} p_j p_l$$

The Christoffel equation can then be written:

$$[\Gamma(\mathbf{p}) - \mathbf{I}] \mathbf{a} = 0$$

This is a **homogeneous linear equation** for $\mathbf{a}$, that can have non-trivial solutions if and only if:

$$\det[\Gamma(\mathbf{p}) - \mathbf{I}] = 0$$

The roots of this equation define the **slowness surface**

Reduction to an eigenvalue problem

Two useful properties

From the definition of the Christoffel matrix, we have:

$$\Gamma(\lambda \mathbf{p}) = \lambda^2 \Gamma(\mathbf{p}),$$

It is a **homogeneous function of order two**

The hyperelastic symmetry implies that the Christoffel matrix is **symmetric**:

$$\Gamma_{ki}(\mathbf{p}) = \frac{1}{\rho} A_{kjil} p_j p_l = \frac{1}{\rho} A_{ilkj} p_j p_l = \frac{1}{\rho} A_{ijkl} p_j p_l = \Gamma_{ik}(\mathbf{p}),$$

Plane waves in a fixed direction

Fixing the orientation of the slowness vector, we can write:

$$\mathbf{p} = \frac{1}{c} \hat{\mathbf{p}},$$

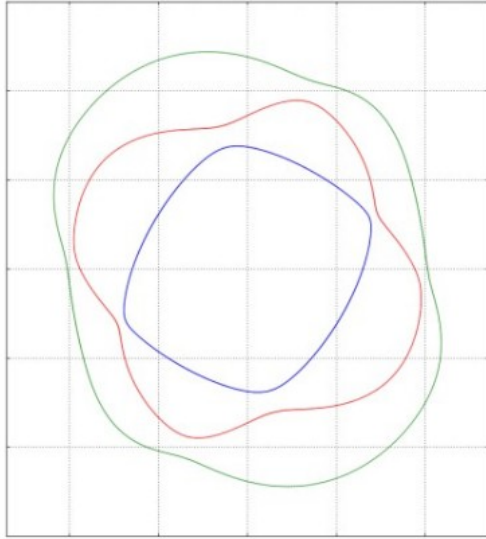
and hence the Christoffel equation reduces to:

$$\Gamma(\hat{\mathbf{p}}) \mathbf{a} = c^2 \mathbf{a}.$$

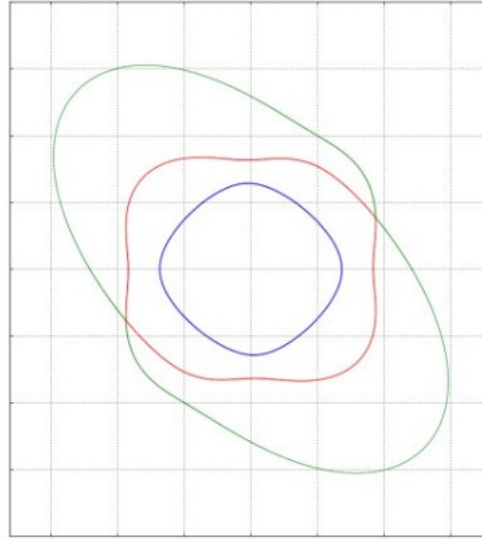
This is a **symmetric eigenvalue problem**:

- 1) Squared phase speeds are **real**, generally **three distinct values**
- 2) Polarisation vectors for distinct squares phase speeds are **orthogonal**
- 3) Degenerate squared phase speeds are possible

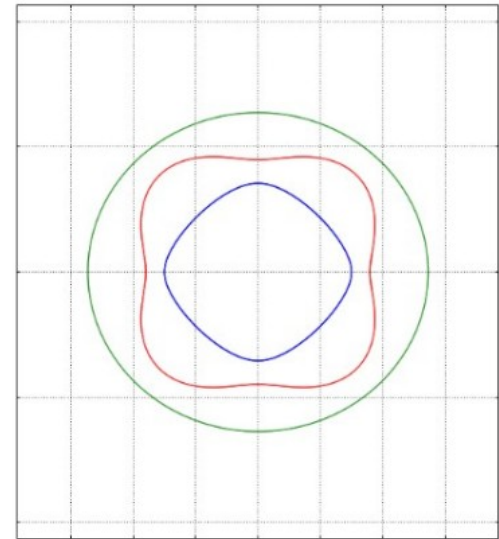
The slowness surface



Three distinct sheets



Two sheets touch



Cubic symmetry

$$[||\mathbf{p}||^2 c_1(\hat{\mathbf{p}})^2 - 1][||\mathbf{p}||^2 c_2(\hat{\mathbf{p}})^2 - 1][||\mathbf{p}||^2 c_3(\hat{\mathbf{p}})^2 - 1] = 0,$$

$$c_1(\hat{\mathbf{p}}) \leq c_2(\hat{\mathbf{p}}) \leq c_3(\hat{\mathbf{p}}),$$

Isotropic media

Isotropic Christoffel matrix

We have seen that the elastic tensor in an isotropic solid takes the form:

$$A_{ijkl} = \lambda \delta_{ij} \delta_{kl} + \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}),$$

Putting into the definition of the Christoffel matrix, we arrive at:

$$\begin{aligned} \Gamma_{ik}(\hat{\mathbf{p}}) &= \frac{\lambda + \mu}{\rho} \hat{p}_i \hat{p}_k + \frac{\mu}{\rho} \delta_{ik}. \\ &= \frac{\lambda + 2\mu}{\rho} \hat{p}_i \hat{p}_k + \frac{\mu}{\rho} (\delta_{ik} - \hat{p}_i \hat{p}_k). \end{aligned}$$

In the final expression, note **orthogonal projections** parallel and perpendicular to the direction of the slowness vector

P-waves

Isotropic Christoffel matrix:

$$\Gamma_{ik}(\hat{\mathbf{p}}) = \frac{\lambda + 2\mu}{\rho} \hat{p}_i \hat{p}_k + \frac{\mu}{\rho} (\delta_{ik} - \hat{p}_i \hat{p}_k).$$

By inspection we identify an eigenvector and eigenvalue:

$$\mathbf{a} \propto \hat{\mathbf{p}} \quad \alpha = \sqrt{\frac{\lambda + 2\mu}{\rho}}.$$

These plane waves are known as **P-waves**:

- Particle motion along the propagation direction
- Phase speed the same in all directions
- Associated with a non-degenerate eigenspace

S-waves

Isotropic Christoffel matrix:

$$\Gamma_{ik}(\hat{\mathbf{p}}) = \frac{\lambda + 2\mu}{\rho} \hat{p}_i \hat{p}_k + \frac{\mu}{\rho} (\delta_{ik} - \hat{p}_i \hat{p}_k).$$

By inspection we identify an eigenvector and eigenvalue:

$$\mathbf{a} \perp \hat{\mathbf{p}} \qquad \beta = \sqrt{\frac{\mu}{\rho}}.$$

These plane waves are known as **S-waves**:

- Particle motion perpendicular to the propagation direction
- Phase speed the same in all directions
- Associated with a two-fold degenerate eigenspace

Slightly anisotropic media

Anisotropy in the Earth



Perturbed eigenvalue problem

We consider a slightly anisotropic elastic tensor:

$$A_{ijkl} = \lambda \delta_{ij} \delta_{kl} + \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}) + s \delta A_{ijkl}$$

Resulting Christoffel matrix:

$$\Gamma_{ik}(\hat{\mathbf{p}}) = \frac{\lambda + \mu}{\rho} \hat{p}_i \hat{p}_k + \frac{\mu}{\rho} \delta_{ik} + s \delta \Gamma_{ik}(\hat{\mathbf{p}}),$$

Eigenvalue problem to be solved:

$$[\mathbf{\Gamma}(\hat{\mathbf{p}}) + s \delta \mathbf{\Gamma}(\hat{\mathbf{p}})] \mathbf{a} = c^2 \mathbf{a}$$

Quasi P-waves

Apply **non-degenerate** perturbation theory:

$$c = \alpha + s \delta\alpha + \dots,$$

$$\mathbf{a} = \hat{\mathbf{p}} + s \delta\mathbf{a} + \dots.$$

First-order term in eigenvalue problem:

$$\Gamma(\hat{\mathbf{p}})\delta\mathbf{a} + \delta\Gamma(\hat{\mathbf{p}})\hat{\mathbf{p}} = \alpha^2\delta\mathbf{a} + 2\alpha\delta\alpha\hat{\mathbf{p}}.$$

Through a standard argument, we arrive at:

$$\delta\alpha = \frac{1}{2\alpha} \hat{\mathbf{p}} \cdot \delta\Gamma(\hat{\mathbf{p}})\hat{\mathbf{p}},$$

Polarisation **almost aligned** with propagation direction, phase speed now **varies with the orientation**

Quasi S-waves

Apply **degenerate** perturbation theory:

$$c = \beta + s \delta\beta + \dots ,$$

$$\mathbf{a} = a_1 \hat{\mathbf{t}}_1 + a_2 \hat{\mathbf{t}}_2 + s \delta\mathbf{a} + \dots ,$$

Leads to 2D eigenvalue problem:

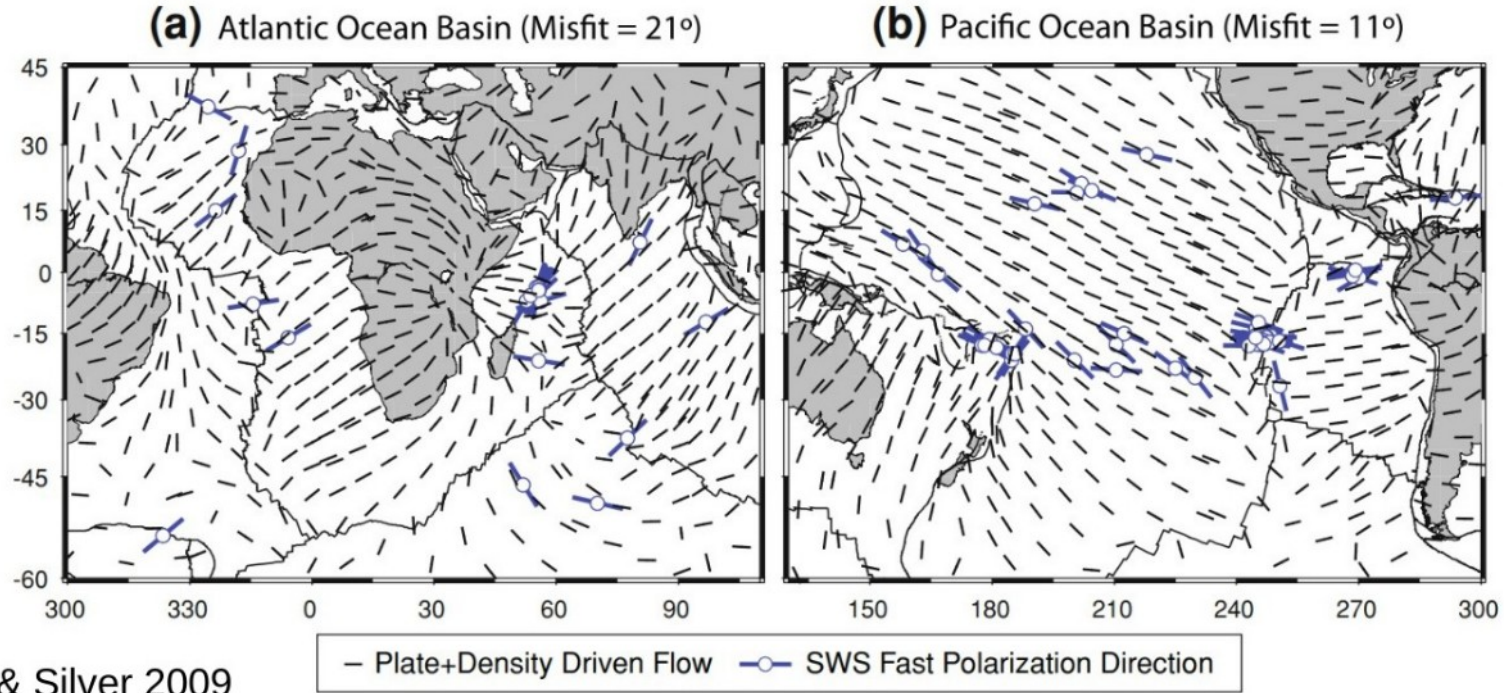
$$\begin{pmatrix} \hat{\mathbf{t}}_1 \cdot \delta\Gamma(\hat{\mathbf{p}}) \hat{\mathbf{t}}_1 & \hat{\mathbf{t}}_1 \cdot \delta\Gamma(\hat{\mathbf{p}}) \hat{\mathbf{t}}_2 \\ \hat{\mathbf{t}}_2 \cdot \delta\Gamma(\hat{\mathbf{p}}) \hat{\mathbf{t}}_1 & \hat{\mathbf{t}}_2 \cdot \delta\Gamma(\hat{\mathbf{p}}) \hat{\mathbf{t}}_2 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = 2\beta\delta\beta \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} .$$

Solution gives **two perturbed phase speeds** which are usually distinct – the initial **degeneracy is split**

Zeroth-order eigenvectors orthogonal to each other and propagation direction, first order correction will have a component along propagation direction

This behaviour is known as **shear wave splitting**

Shear wave splitting and mantle flow



Convection in the mantle creates large-scale **alignment** of anisotropic mineral grains with the **flow direction** – produces an effective seismic anisotropy. Example here shows the relation between shallow anisotropy and plate motion

Interaction of plane waves with boundaries

Topic covered quantitatively in previous years but never examined

Quantitative details now removed from the course for reasons of time

The usual **reflections** and **transmissions** of waves occur when a plane elastic wave interacts with a boundary.

The only interesting feature is the **conversion** between different wave types. For example, in an isotropic medium, an incident P-wave can produce reflected and transmitted P- and S- waves

